

Law of Total Probability; Bayes' Theorem

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MATH 241

Lecture #5

Reminders!

- **Homework #2** is due on Friday, 9/18 at 11:59pm
 - ▶ Probability problems plus an R problem
- **Checkpoint #2** at end of class!
- **Extra office hours:** tomorrow from 10:30am-11:30am in Rocky 106A

Learning Objectives

- **Visualize** and **interpret** concepts of conditional probability, through the use of formulas and Bayes' theorem
- **Investigate** the Law of Total Probability, and how it connects to identifying conditional probabilities
- **Calculate** conditional probabilities

Formulas to Keep in Mind

Conditional Probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Multiplication Rule:

$$P(A \cap B) = P(B|A) \times P(A)$$

Law of Total Probability:

$$P(B) = \sum_{j=1}^n P(B|A_j)P(A_j)$$

Bayes' Theorem:

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

Example #2: Multiplication Rule Example with $n = 3$

For example, let E_1 be the event that it is sunny and warm, E_2 be the event that Joe swims outdoors, and E_3 be the event that Joe gets sunburned. Then what is the probability that it's sunny and warm and Joe swims outdoors and Joe gets sunburned?

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$$\begin{aligned}P(E_1 E_2 E_3) &= P(E_1)P(E_2|E_1)P(E_3|E_1, E_2) \\ &= P(E_2)P(E_3|E_2)P(E_1|E_3, E_2)\end{aligned}$$

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In this context, it is most logical to use this ordering:

$$P(E_1 E_2 E_3) = P(E_1)P(E_2|E_1)P(E_3|E_1, E_2)$$

Law of Total Probability

If $A_1, \dots, A_n$ form a partition of the sample space S with $P(A_i) > 0$ for all i , then:

$$P(B) = \sum_{j=1}^n P(B|A_j)P(A_j)$$

Ex: If B was the event of getting a damaged product from Amazon, and A_i refers to the shipping carrier used, then the Law of Total Probability:

- Takes the probability of a damaged good given the carrier and multiplies by the chance the carrier is being used
- Sums up each of these joint probabilities

Proof

Proof.

Let A and B be events. We can rewrite B as the following:

$$B = (BA_1) \cup \cdots \cup (BA_n)$$

Since $BA_1, \dots, BA_n$ are mutually exclusive, then by axiom #2 of probability:

$$\begin{aligned} P(B) &= P(BA_1) + \cdots + P(BA_n) \\ \implies P(B) &= P(B|A_1)P(A_1) + \cdots + P(B|A_n)P(A_n) \end{aligned}$$



Example #1: Dice Rolls

Suppose I roll two fair dice and am curious about the following events:

- A : at least one dice roll is a 3
- B_1 : sum of die is 8
- B_2 : sum of die is less than 8
- B_3 : sum of die is greater than 8

Find $P(A)$ using the law of total probability.

Example #1: Dice Rolls

- A : at least one dice roll is a 3
- B_1 : sum of die is 8
- B_2 : sum of die is less than 8
- B_3 : sum of die is greater than 8

We can decompose $P(A)$ as the following:

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3)$$

We have already found $P(A|B_1)$ and $P(B)$ from the earlier examples. For the other quantities:

$$P(B_2) = \frac{21}{36}$$
$$P(A|B_2) = \frac{7}{21}$$

$$P(B_3) = \frac{10}{36}$$
$$P(A|B_3) = \frac{2}{10}$$

Example #1: Dice Rolls

Then,

$$\begin{aligned}P(A) &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) \\&= \frac{2}{5} \times \frac{5}{36} + \frac{8}{21} \times \frac{21}{36} + \frac{2}{10} \times \frac{10}{36} \\&= \frac{12}{36} \\&= \frac{1}{3}\end{aligned}$$

Diagnostic Testing

A certain medical test for a disease is highly accurate in that if a person has disease, then the test gives a positive result 84% of the time and if a person doesn't have disease, then the test gives a positive result only 1% of the time. Data indicate that 1 in 1000 people in a community have the disease.

Question: Suppose you take the test and receive a positive response. What's the probability you have the disease?

Note: You will use all three rules discussed earlier in the week.

Bayes' Theorem

What you might have discovered when expanding out the conditional probability. We can find the probability of some event A given event B if we know the conditional probabilities of B given events A and A^C individually (as well as probabilities of events A and A^C), as seen below:

$$P(A|B) = \frac{P(A, B)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^C)P(A^C)}$$

- Numerator is a result of the multiplication rule
- Denominator is a result of the law of total probability

Example: Diagnostic Testing

Let D be the event of a person having the disease and $+$ be the event of getting a positive result. Then,

$$\begin{aligned}P(+|D) &= 0.84 \\P(+|D^C) &= 0.01\end{aligned}\qquad P(D) = 0.0001$$

Using the definition of conditional probability, which can be transformed into Bayes' Theorem, we get:

$$\begin{aligned}P(D|+) &= \frac{P(D, +)}{P(+)} \\&= \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|D^C)P(D^C)} \\&= \frac{0.84 \times 0.001}{0.84 \times 0.001 + 0.01 \times 0.999} \\&\approx 0.0776\end{aligned}$$

Bayes' Theorem (General)

Question: Suppose we can make a partition of S using $A_1, A_2, \dots, A_n$. How can Bayes' Theorem account for this?

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$$P(A_j|B) = \frac{P(A_j, B)}{P(B)} = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

Activity #1: Cookies

It's autumn, and it's time to bake some cookies! Denote A as the event of using a recipe for chocolate-chip cookies, but denote B_1 , B_2 , and B_3 as the event of good, bad, and okay cookies being baked. After years of baking, I find that the probabilities are:

- $P(A) = 0.7$
- $P(A \cap B_1) = 0.42$
- $P(B_1) = 0.5$
- $P(A \cap B_2) = 0.07$
- $P(B_2) = 0.3$
- $P(B_3) = 0.2$

Use Bayes' Theorem to find the probability of okay cookies being baked with a recipe. Note the use of the multiplication rule here.

The Axioms on Conditional Probabilities

The axioms of probability also apply to conditional probabilities. Treated on some event E ,

$$P(S|E) = 1$$

$$P(\emptyset|E) = 0$$

$$A_1, \dots, A_n \text{ disjoint} \implies P\left(\bigcup_{i=1}^n A_i|E\right) = \sum_{i=1}^n P(A_i|E)$$

$$P(A^C|E) = 1 - P(A|E)$$

$$P(A \cup B|E) = P(A|E) + P(B|E) - P(A \cap B|E)$$

Independence

- We see that conditioning on an event can change the probability of another event
 - ▶ e.g. probability of passing exam depends on whether you study or not
- Suppose if one event does not influence another event occurring
 - ▶ e.g. probability of rolling a 4 on the first dice roll does not affect the probability of rolling a 6 on the next dice roll
 - ▶ This brings up independence!

Checkpoint!

- You may use handwritten or typed notes on paper for this checkpoint. No electronics are permitted.
- Read and follow all directions; points will be deducted if not all work is shown.
- You have until the rest of class to finish the questions.

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